



# Using Digital Trace Data to Differentiate Emigration from Return Migration of Scholars

Faeze Ghorbanpour<sup>1</sup>, Thiago Zordan Malaguth<sup>2</sup>, Aliakbar Akbaritabar<sup>2</sup>

<sup>1</sup>Ludwig Maximilian University of Munich

<sup>2</sup>Max Planck Institute for Demographic Research  
akbaritabar@demogr.mpg.de

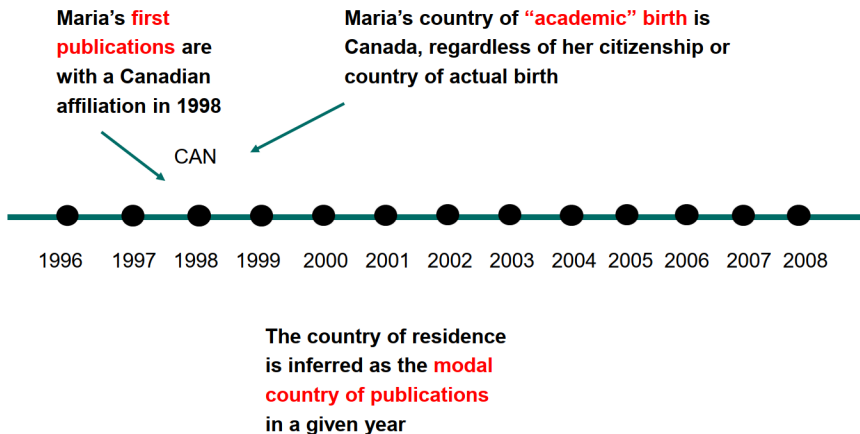
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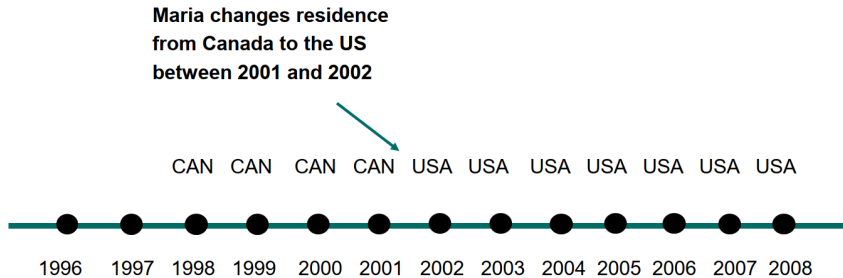
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- ▶ In the context of migration, when we have already observed a consistent migration event using digital traces:
  - ▶ Identifying nationality enables differentiation between emigration and return migration.

# Identifying migration events (1)



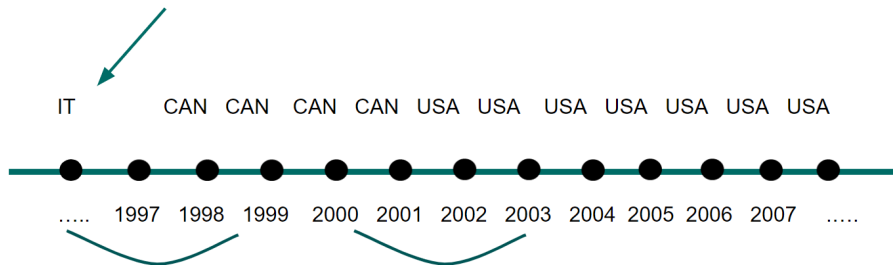
# Identifying migration events (2)



**Maria's modal countries  
of publications over time**

# How the ML method works

Maria's "full name", "**Maria Casnici**" allows a rough guess of her nationality



Maria's **migration history** can change!

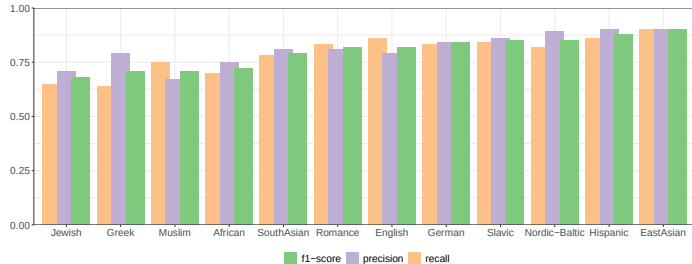


- ▶ We detect the nationality with the least available data, full names, using Machine Learning with fasttext and char CNN.
- ▶ We use it instead of the country of academic origin in studying the migration of scholars.
- ▶ We exclude the main countries of immigration with a high diversity of scholars such as the US, the UK, Canada, Australia and South Africa from our training, test and evaluation steps.
- ▶ Once our model shows good performance, we use these data to analyze the composition of scholars' population and return migration.

- Nested structure of regions and countries of origin based on the name in three levels (left) and performance of our Machine Learning model in level 1 in the most inner circle (right).



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# Training, evaluation and analysis data (1/3)

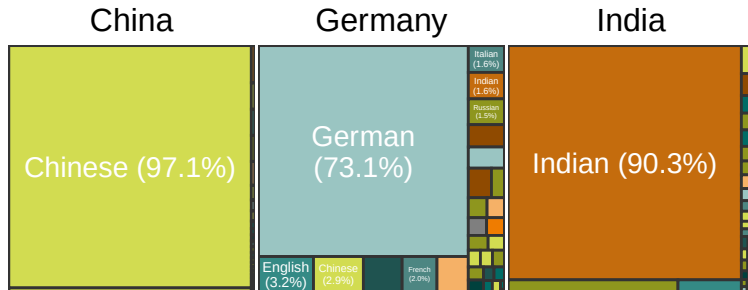


|                                      | Name                                                                                         | Description                                                                                                      | Volume    |
|--------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------|
| Training data                        | Wikipedia notable people's mobility                                                          | This dataset contains mobility data of notable people whose biography is registered in Wikipedia.                | 2,190,457 |
|                                      | Nationality Estimate                                                                         | This dataset is constructed with name-nationality pairs by parsing the English-language Wikipedia.               | 696,136   |
|                                      | WikiBio (Wikipedia Biography Dataset)                                                        | This dataset gathers biographies from English Wikipedia.                                                         | 418,978   |
|                                      | Wikipedia Person Dataset                                                                     | This dataset gathers the infobox with the description of the person's page.                                      | 355,468   |
|                                      | Politicians on Wikipedia                                                                     | This dataset contains information about politicians from Wikipedia.                                              | 54,877    |
|                                      | Pantheon 1.0 dataset                                                                         | This dataset includes the biographies present in more than 25 languages in Wikipedia.                            | 16,071    |
|                                      | Wikidata                                                                                     | Processed by our colleague.                                                                                      | 131,731   |
| <b>SUM</b>                           |                                                                                              |                                                                                                                  | 3,874,776 |
| <b>SUM after removing duplicates</b> |                                                                                              |                                                                                                                  | 2,634,445 |
| Test data                            | Athletes                                                                                     | Data collected from the sports sites.                                                                            | 42,413    |
|                                      | International Union for the Scientific Study of Population (IUSSP) public members' directory | Information on IUSSP member's self-reported nationality, field of specialty, spoken languages, affiliation, etc. | 1,547     |

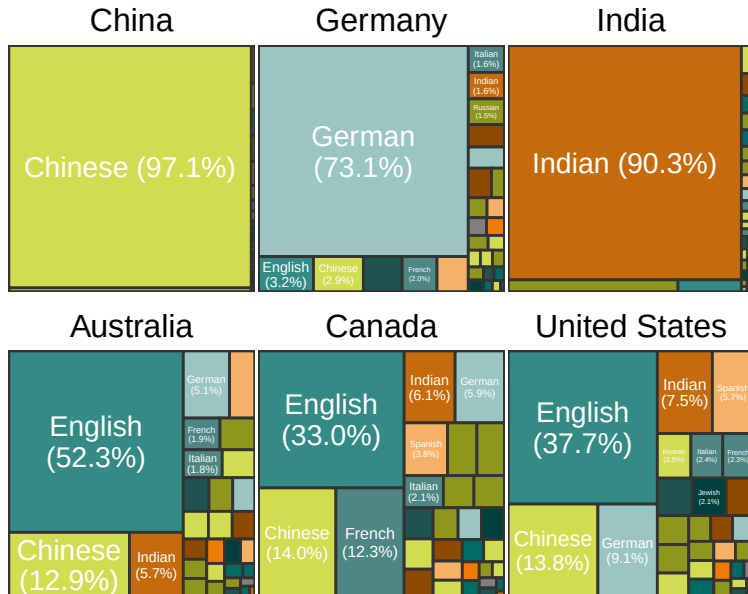
|            | Level 1        |                |                | Level 2        |                |                | Level 3        |                |                |
|------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
|            | F1-macro       | F1-weighted    | Accuracy       | F1-macro       | F1-weighted    | Accuracy       | F1-macro       | F1-weighted    | Accuracy       |
| Char LSTM  | 0.24191        | 0.37892        | 0.27712        | 0.054929       | 0.151025       | 0.30124        | 0.03118        | 0.05980        | 0.16536        |
| Char CNN   | 0.36432        | 0.53929        | 0.53929        | 0.53929        | 0.53929        | 0.53929        | 0.18906        | 0.06785        | 0.18906        |
| Fasttext   | <b>0.80887</b> | 0.83598        | 0.83619        | 0.69132        | 0.78487        | 0.78956        | 0.33205        | 0.63445        | 0.65139        |
| ByT5 small | 0.74566        | <b>0.83854</b> | <b>0.83759</b> | <b>0.72737</b> | 0.80971        | <b>0.81121</b> | 0.41666        | <b>0.66789</b> | <b>0.68356</b> |
| ByT5 base  | 0.74247        | 0.83557        | 0.83453        | 0.72491        | <b>0.80673</b> | 0.80828        | <b>0.42044</b> | 0.66453        | 0.67672        |
| ByT5 large | 0.79765        | 0.82987        | 0.82944        | 0.72051        | 0.80209        | 0.80349        | 0.41723        | 0.65799        | 0.66822        |

|          | Level 1  |             |          | Level 2  |             |          | Level 3  |             |          |
|----------|----------|-------------|----------|----------|-------------|----------|----------|-------------|----------|
|          | F1-macro | F1-weighted | Accuracy | F1-macro | F1-weighted | Accuracy | F1-macro | F1-weighted | Accuracy |
| Athletes | 0.79276  | 0.83423     | 0.83222  | 0.75739  | 0.79889     | 0.79889  | 0.40916  | 0.66433     | 0.66003  |
| IUSSP    | 0.71360  | 0.84669     | 0.84223  | 0.74597  | 0.82243     | 0.82119  | 0.41729  | 0.66523     | 0.66262  |

# Population's Composition

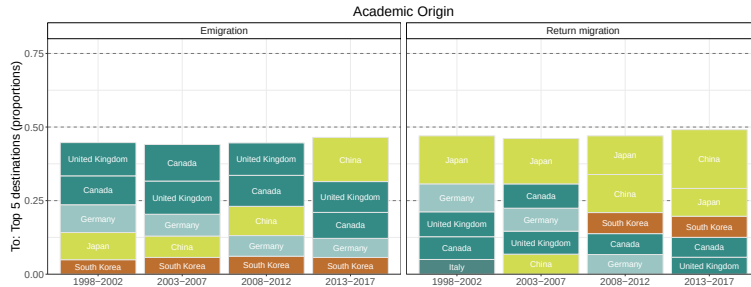


# Population's Composition

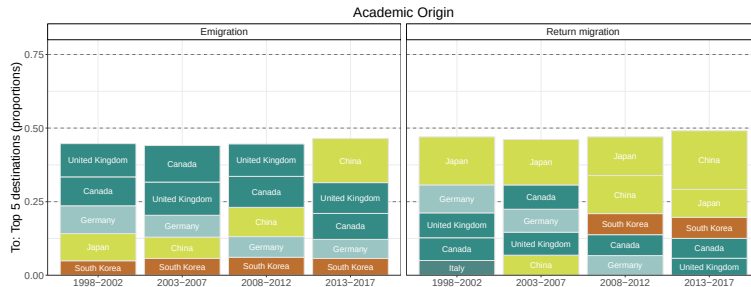
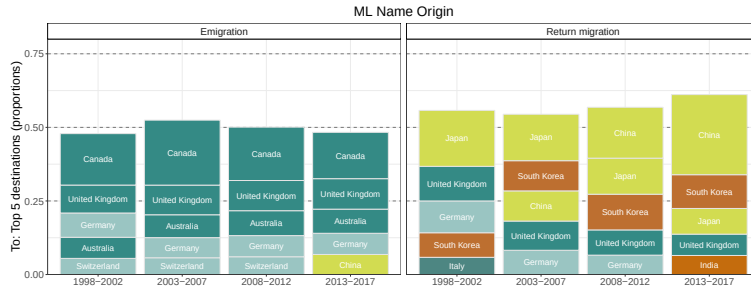




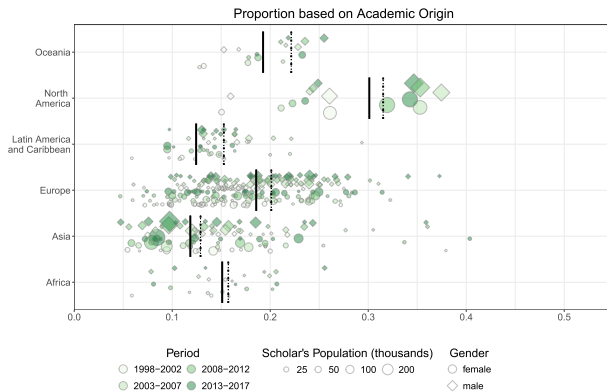
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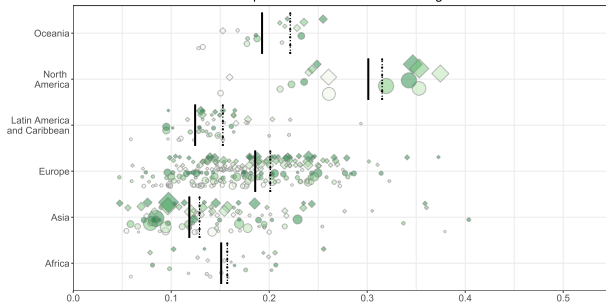


# Proportion of return migration

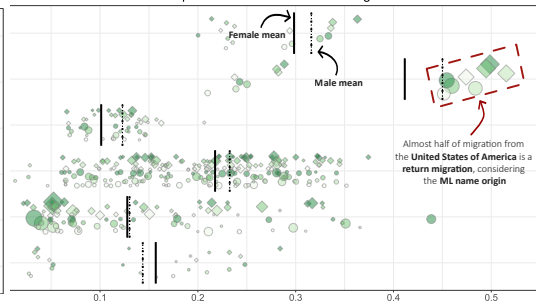


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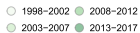
Proportion based on Academic Origin



Proportion based on ML Name Origin



Period



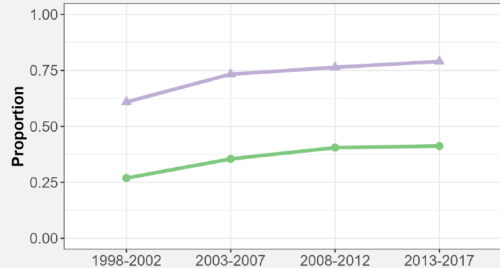
Scholar's Population (thousands)



Gender



In the most recent period, 79% of migrants from the US to China are classified as return migrants using the ML-assigned name origin



Proportion of migration from US to China classified as return migration based on ML name origin and Academic Origin

# Takeaways!



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- ▶ In some migration corridors, our model's results are the most reliable.

# Thanks for your attention

Questions and comments are welcome!



Aliakbar Akbaritabar

Research scientist

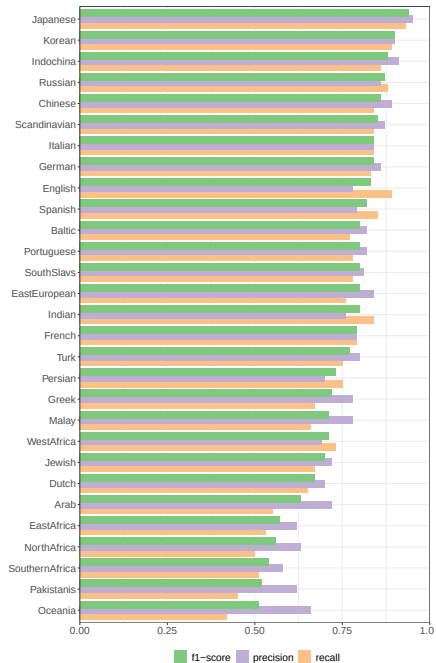
Deputy Head for Training

Max Planck Institute for Demographic Research  
(MPIDR), Rostock, Germany.

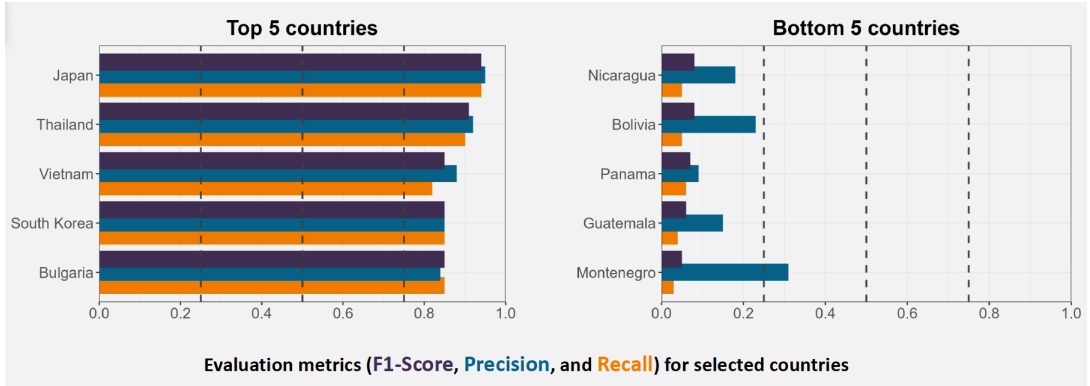
Email: [akbaritabar@demogr.mpg.de](mailto:akbaritabar@demogr.mpg.de)

Twitter/BlueSky/Mastodon: @akbaritabar

# Evaluation at level 2



# Evaluation at level 3 - countries



# Examples from IUSSP's members directory



|                                      | Level 1       |               | Level 2      |           | Level 3        |                |
|--------------------------------------|---------------|---------------|--------------|-----------|----------------|----------------|
| Name                                 | Reported      | Predicted     | Reported     | Predicted | Reported       | Predicted      |
| <b>Mohammad Jalal Abbasi-Shavazi</b> | Muslim        | Muslim        | Persian      | Persian   | Iran           | Iran           |
| <b>Emilio Zagheni</b>                | Romance       | Romance       | Italian      | Italian   | Italy          | Italy          |
| <b>Guy Abel</b>                      | English       | Romance       | English      | French    | United Kingdom | France         |
| <b>Mikko Myrskylä</b>                | Nordic-Baltic | Nordic-Baltic | Nordic       | Nordic    | Finland        | Finland        |
| <b>Vladimir Canudas-Romo</b>         | Hispanic      | Slavic        | Spanish      | Russian   | Mexico         | Russia         |
| <b>Midanna de Almada</b>             | African       | Hispanic      | South Africa | Spanish   | Namibia        | Portugal       |
| <b>Preeti Upadhyay</b>               | South Asia    | South Asia    | Indian       | Indian    | India          | India          |
| <b>Louis Lohle-Tart</b>              | German        | Romance       | Dutch        | French    | Belgium        | France         |
| <b>Nico Van Nimwegen</b>             | German        | German        | Dutch        | Dutch     | Netherlands    | Netherlands    |
| <b>Andrew "Amos" Channon</b>         | English       | English       | English      | English   | United Kingdom | United Kingdom |